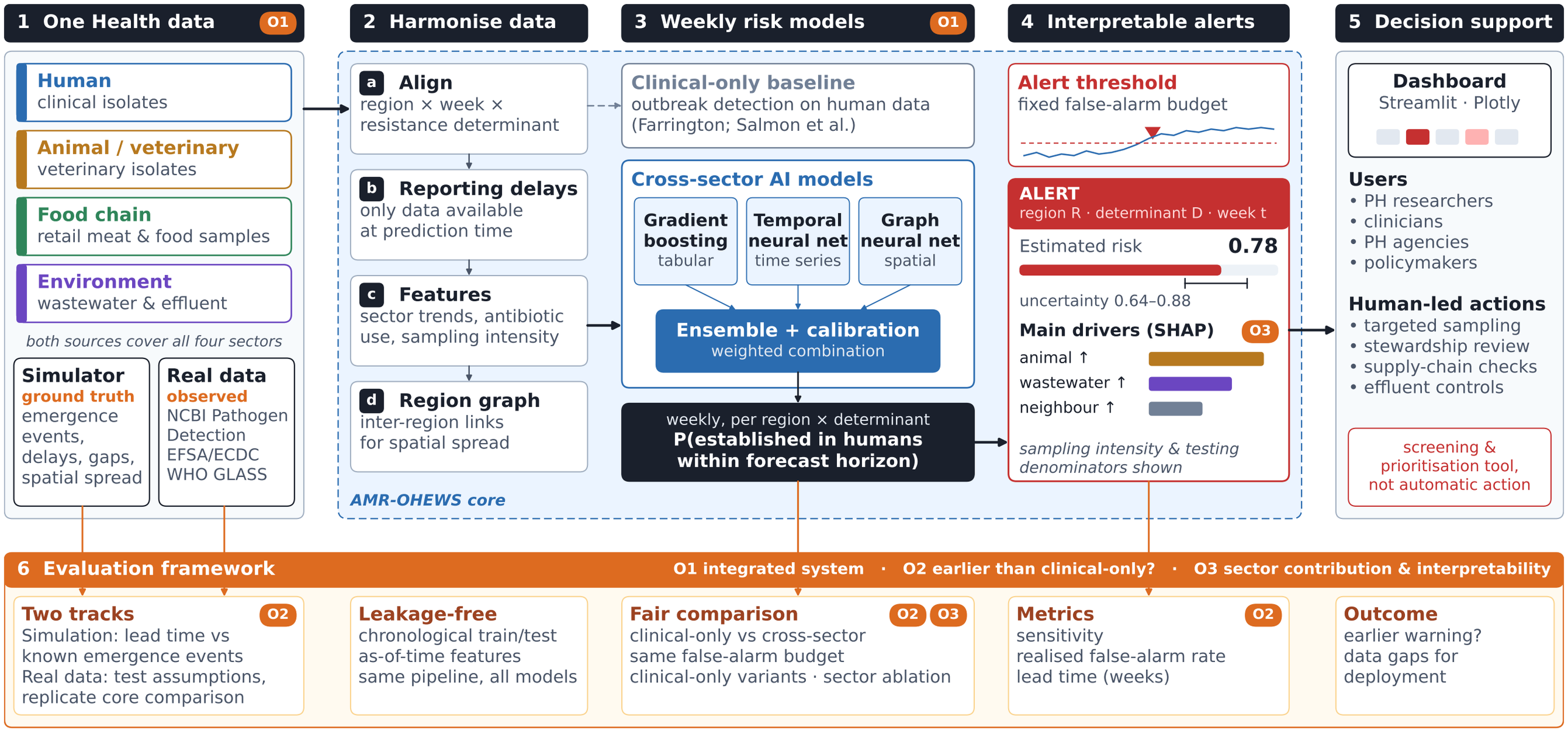

1 One Health data
O1
Human
clinical isolates
Animal / veterinary
veterinary isolates
Food chain
retail meat & food samples
Environment
wastewater & effluent
both sources cover all four sectors
Simulator
ground truth
emergence events, delays, gaps, spatial spread
Real data
observed
NCBI Pathogen Detection
EFSA/ECDC
WHO GLASS
2 Harmonise data
a Align
region × week × resistance determinant
b Reporting delays
only data available at prediction time
c Features
sector trends, antibiotic use, sampling intensity
d Region graph
inter-region links for spatial spread
AMR-OHEWS core
3 Weekly risk models
O1
Clinical-only baseline
outbreak detection on human data (Farrington; Salmon et al.)
Cross-sector AI models
Gradient boosting
tabular
Temporal neural net
time series
Graph neural net
spatial
Ensemble + calibration
weighted combination
weekly, per region × determinant
P(established in humans within forecast horizon)
4 Interpretable alerts
Alert threshold
fixed false-alarm budget
ALERT
region R · determinant D · week t
Estimated risk
0.78
uncertainty 0.64–0.88
Main drivers (SHAP)
O3
animal ↑
wastewater ↑
neighbour ↑
sampling intensity & testing denominators shown
5 Decision support
Dashboard
Streamlit · Plotly
Users
• PH researchers
• clinicians
• PH agencies
• policymakers
Human-led actions
• targeted sampling
• stewardship review
• supply-chain checks
• effluent controls
screening & prioritisation tool, not automatic action
6 Evaluation framework
O1 integrated system · O2 earlier than clinical-only? · O3 sector contribution & interpretability
Two tracks
O2
Simulation: lead time vs known emergence events
Real data: test assumptions, replicate core comparison
Leakage-free
chronological train/test
as-of-time features
same pipeline, all models
Fair comparison
O2 O3
clinical-only vs cross-sector
same false-alarm budget
clinical-only variants · sector ablation
Metrics
O2
sensitivity
realised false-alarm rate
lead time (weeks)
Outcome
earlier warning?
data gaps for deployment